

Sales Dashboard Analytics using Excel

1. Project Objective

The objective of this project is to analyse sales data and create an interactive dashboard using Microsoft Excel. The dashboard helps visualize sales performance across product categories, regions, payment methods, store types, and years to support business decision-making.

2. Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Data Cleaning & Transformation Techniques

3. Dataset Overview

The dataset contains sales transaction information including:

- Product Category
- Region
- Payment Type
- Store Type
- Sales Revenue
- Units Sold
- Year of Sale

4. Data Cleaning & Transformation

The following steps were performed:

- Removed duplicate records.
- Checked for missing values and corrected inconsistencies.
- Standardized data formats.
- Created Pivot Tables for data aggregation.
- Applied filters and slicers for dynamic analysis.

Data Cleaning and Transformation Screenshots:

Customer_ID	Customer_ID
C-1464	Cust1464
C-1486	Cust1486
C-1491	Cust1491
C-1499	Cust1499

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA	
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	
CUST1037	Christopher Carter	27	Female	Brendastad	South Dakota	USA	
Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
Cust1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Unknown
Cust1018	Rhonda Moss	43	Male	South William	Kansas	USA	Unknown
Cust1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	Unknown
Cust1037	Christopher Carter	27	Female	Brendastad	South Dakota	USA	Unknown

Name	Name
sheena bonilla	Sheena Bonilla
Eugene Nelson	Eugene Nelson
dr. Benjamin Price	Dr. Benjamin Price
jeremy harris	Jeremy Harris

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	198.4164
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	228.7893
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	2742.15
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	0.16	COD	774.0684
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	745.3	0.18	Credit Card	1833.438
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	0.19	COD	357.2343
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	1335.027
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	1054.5612
SALE8	2025-05-04	CUST1187	PROD99	STORE5		448.25	0.07	PayPal	1250.6175
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	1344.6818
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	259.1259
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	952.5744
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	2552.5368
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	316.9	0.3	PayPal	443.66
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1145.8548
SALE15	2024-12-10	CUST1435	PROD36	STORE9		871.34	0.04	COD	2509.4592
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	677.97
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	774.1426
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	0.25	COD	838.845
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4		0.04	COD	1782.4512

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	\$ 245.98	0.09	Credit Card	\$ 245.07
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	\$ 403.71	0.27	PayPal	\$ 1,210.40
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	\$ 43.29	0.25	Credit Card	\$ 172.41
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	\$ 392.89	0.16	COD	\$ 392.05
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	\$ 17.31	0.18	Credit Card	\$ 51.11
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	\$ 465.34	0.19	COD	\$ 1,395.21
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	\$ 461.47	0.15	PayPal	\$ 922.09
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	\$ 138.05	0.23	PayPal	\$ 551.43
SALE8	04-05-2025	CUST1187	PROD99	STORE5	1	\$ 201.41	0.07	PayPal	\$ 200.48
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	\$ 179.77	0.23	PayPal	\$ 358.77
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	\$ 166.06	0.07	Credit Card	\$ 165.13
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	\$ 326.06	0.17	Credit Card	\$ 1,303.41
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	\$ 131.24	0.01	Credit Card	\$ 523.97
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	\$ 419.28	0.3	PayPal	\$ 837.86
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	\$ 166.06	0.29	PayPal	\$ 663.53
SALE15	10-12-2024	CUST1435	PROD36	STORE9	1	\$ 223.43	0.04	COD	\$ 222.47
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	\$ 240.63	0.25	Credit Card	\$ 480.51
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	\$ 254.04	0.23	Credit Card	\$ 507.32
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	\$ 326.06	0.25	COD	\$ 651.37

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
P-16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	443.40	0
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	480
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowmar	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	254.04	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	254.04	352
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	0
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	254.04	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268

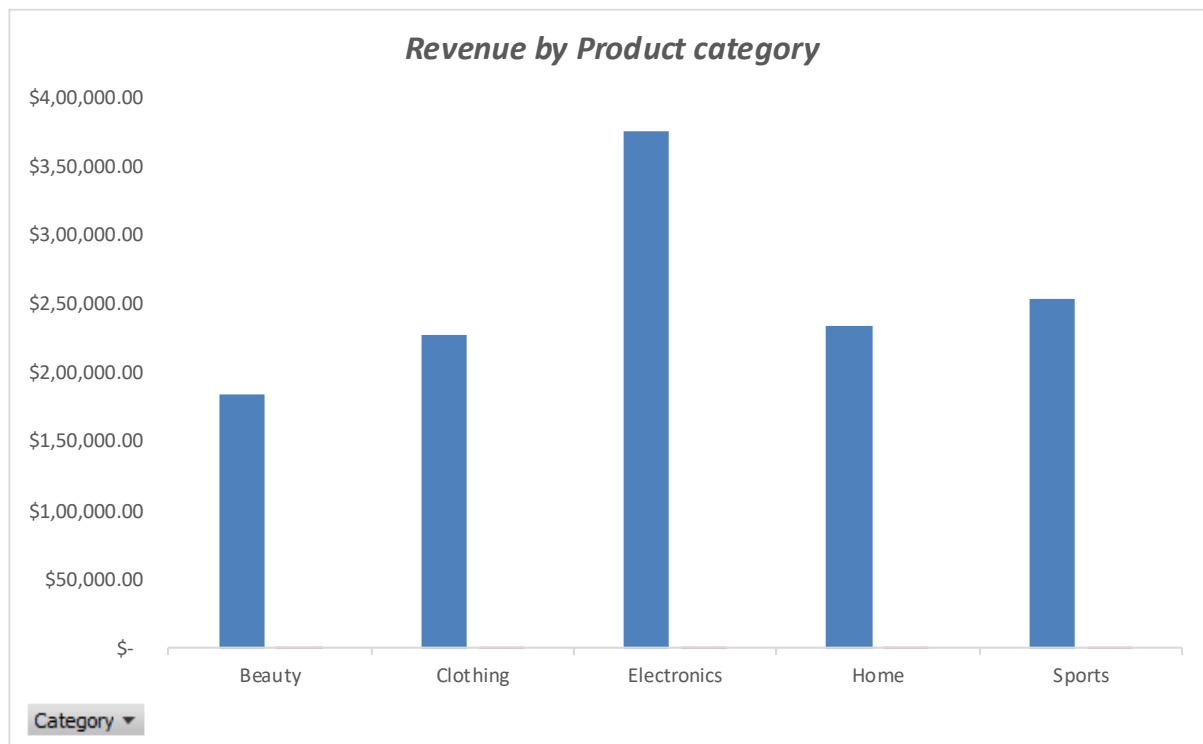
Product_ID	Product_Name	Category	Sub_Category	Brand	Product Brand	Cost	Stock
Prod0	Brother Laser Printer	Electronics	Computer	Hunt Inc	Hunt Inc Computer	112.42	416
Prod1	Men'S Graphic T-Shirt	Clothing	Tops	Terry-Moore	Terry-Moore Tops	342.56	181
Prod2	Women'S Casual Hoodie	Clothing	Outerwear	Sims Inc	Sims Inc Outerwear	254.04	352
Prod3	Men'S Running Jacket	Clothing	Outerwear	Pineda Ltd	Pineda Ltd Outerwear	364.67	452
Prod4	Women'S Yoga Pants	Clothing	Bottoms	Gonzales-Davis	Gonzales-Davis Bottoms	306.46	487
Prod5	Men'S Leather Belt	Clothing	Accessories	Baldwin, Garza And Gonz	Baldwin, Garza And Gonz	261.00	59
Prod6	Herbal Face Moisturizer	Beauty	Skincare	Davidson Plc	Davidson Plc Skincare	312.80	350
Prod7	Led Table Lamp	Home	Lighting	Harrell Llc	Harrell Llc Lighting	218.86	170
Prod8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	Perez-Johnston Audio	439.07	142
Prod9	Adjustable Dumbbell Set	Sports	Fitness	Hill Plc	Hill Plc Fitness	495.72	323
Prod10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	Hamilton Group Kitchen	298.86	308
Prod11	Hd Smart Tv Stick	Electronics	Accessories	Gibson, Hobbs And Weave	Gibson, Hobbs And Weave	443.40	0
Prod12	Hair Strengthening Shampoo	Beauty	Haircare	Rice Llc	Rice Llc Haircare	115.82	480
Prod13	Usb-C Fast Charger	Electronics	Accessories	Dawson-Perkins	Dawson-Perkins Accessori	242.23	453
Prod14	Decorative Wall Mirror	Home	Decor	Luna, Anderson And Bowr	Luna, Anderson And Bowr	48.01	439
Prod15	4K Home Projector	Electronics	Appliances	Thomas Llc	Thomas Llc Appliances	465.34	301
Prod16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	Erickson-Green Grooming	337.42	256
Prod17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy And Wilson	Ford, Murphy And Wilson A	144.53	367
Prod18	Women'S Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	Guerra-Hendrix Bottoms	254.04	343
Prod19	Knee Support Brace	Sports	Equipment	May-Reese	May-Reese Equipment	21.34	65
Prod20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	Soto-Ellis Accessories	131.51	0
Prod21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	Gardner-Montes Storage	123.46	76
Prod22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda And Rodrig	Flores, Pineda And Rodrig	484.62	432
Prod23	Men'S Formal Shirt	Clothing	Tops	Moon, Mora And Santos	Moon, Mora And Santos Tr	254.04	360
Prod24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	Cook Ltd Outdoor	166.06	340
Prod25	Women'S Scarf Set	Clothing	Accessories	Brown And Sons	Brown And Sons Accessori	174.02	268
Prod26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	Reed-Schmidt Audio	322.41	213
Prod27	Decorative Floor Vase	Home	Decor	Khan, Jones And Paul	Khan, Jones And Paul Dec	254.04	378

6. Dashboard Development

An interactive dashboard was developed to analyse:

Revenue by Product Category:

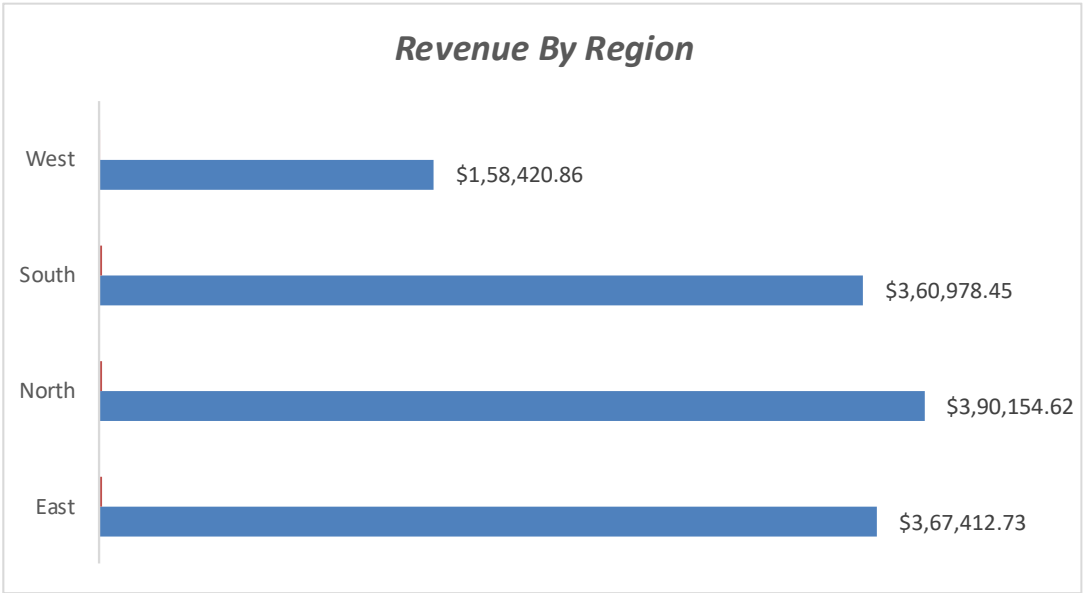
<i>Revenue by Product category</i>		
Product Category ▼	Total Revenue	Total Units
Beauty	\$ 1,84,090.34	738
Clothing	\$ 2,27,607.42	844
Electronics	\$ 3,76,365.51	1165
Home	\$ 2,35,096.63	965
Sports	\$ 2,53,806.77	1241
Grand Total	\$ 12,76,966.67	4953



- Compared revenue generated across different product categories.
- Identified the highest and lowest performing product categories.
- Analysed category-wise contribution to total sales.

Revenue by Region:

Revenue By Region		
Region	Total Revenue	Total units
East	\$ 3,67,412.73	1447
North	\$ 3,90,154.62	1499
South	\$ 3,60,978.45	1414
West	\$ 1,58,420.86	593
Grand Total	\$ 12,76,966.67	4953

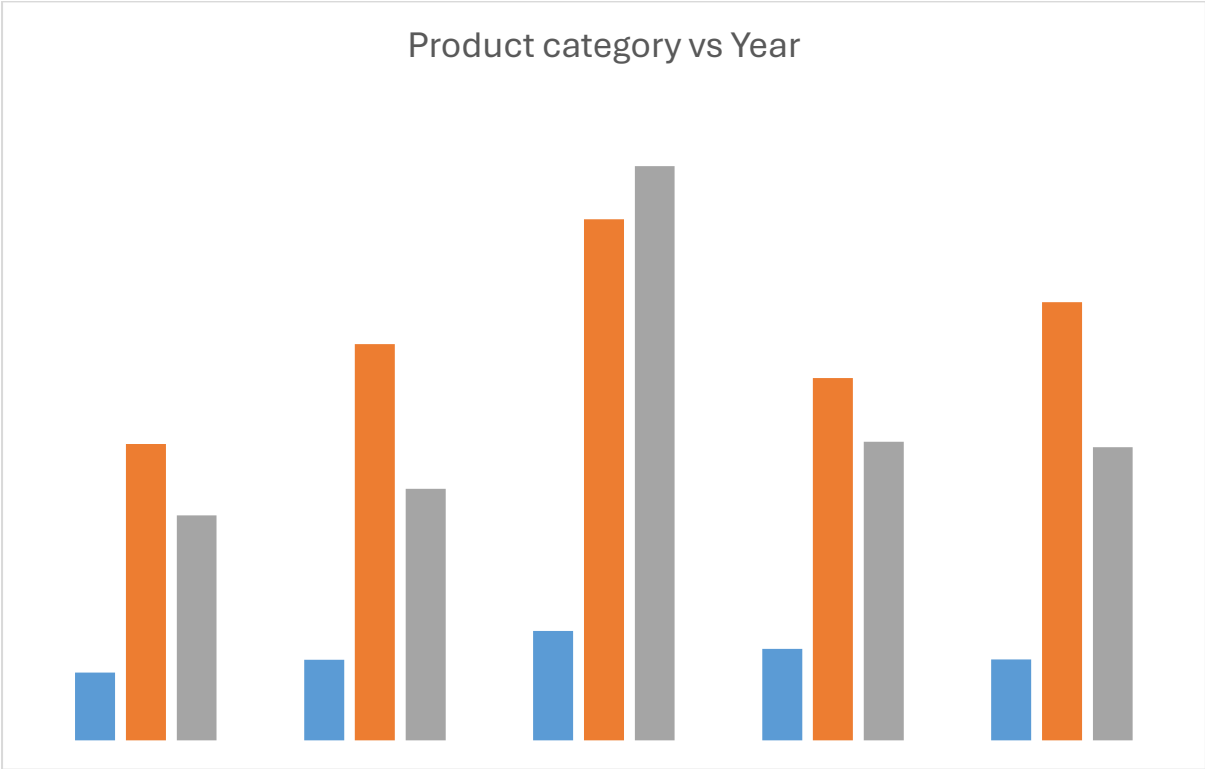


- Evaluated sales performance across different regions.
- Identified regions generating the highest revenue.
- Compared regional sales trends for business insights.

Product Category vs Year Performance:

Product category vs Year

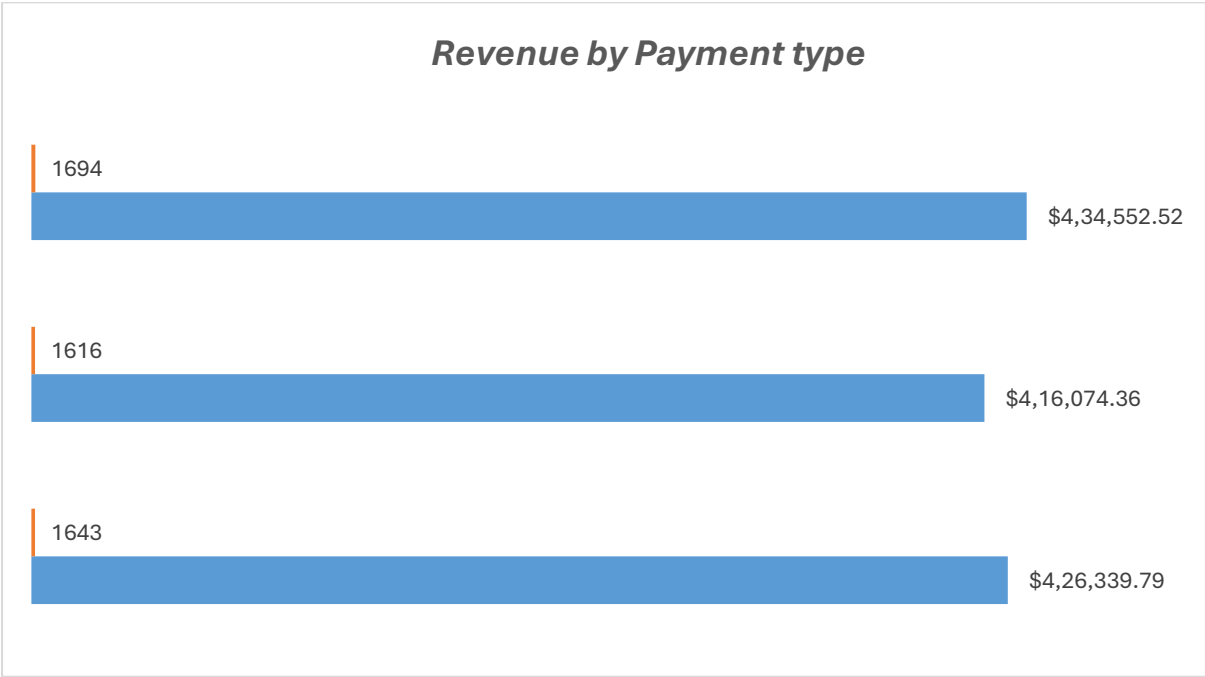
Sum of Total_Amount	Year of sale			
Product category	2023	2024	2025	Grand Total
Beauty	\$ 21,205.57	\$ 92,578.38	\$ 70,306.39	\$ 1,84,090.34
Clothing	\$ 25,229.48	\$1,23,798.72	\$ 78,579.22	\$ 2,27,607.42
Electronics	\$ 34,176.09	\$1,62,793.12	\$1,79,396.30	\$ 3,76,365.51
Home	\$ 28,630.17	\$1,13,186.17	\$ 93,280.30	\$ 2,35,096.63
Sports	\$ 25,254.29	\$1,36,899.40	\$ 91,653.07	\$ 2,53,806.77
Grand Total	\$ 1,34,495.60	\$6,29,255.79	\$5,13,215.27	\$ 12,76,966.67



- Compared yearly revenue across product categories.
- Identified categories with consistent growth.
- Analysed category performance trends over different years.

Revenue by Payment Type:

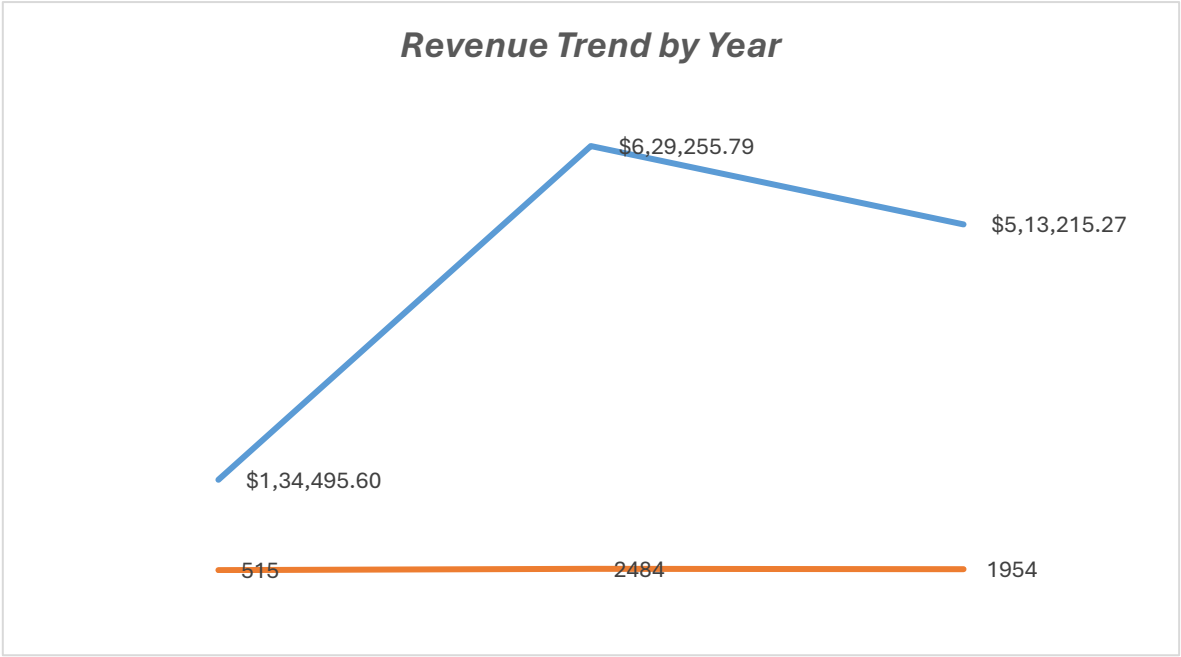
Revenue by Payment type		
Payment Type	Total Revenue	Total units
Cod	\$ 4,26,339.79	1643
Credit Card	\$ 4,16,074.36	1616
Paypal	\$ 4,34,552.52	1694
Grand Total	\$ 12,76,966.67	4953



- Analyzed revenue generated through different payment methods.
- Identified the most preferred payment option among customers.
- Compared payment-wise sales contribution.

Revenue Trend by Year:

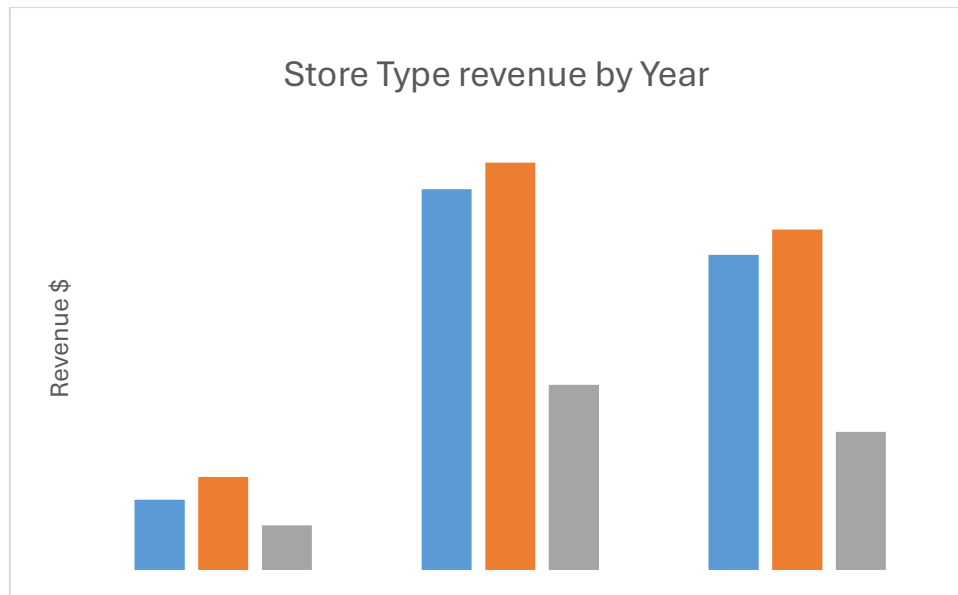
Revenue Trend by Year		
Year of sale	Total Revenue	Total Units
2023	\$ 1,34,495.60	515
2024	\$ 6,29,255.79	2484
2025	\$ 5,13,215.27	1954
Grand Total	\$ 12,76,966.67	4953



- Examined revenue growth over multiple years.
- Identified upward and downward sales trends.
- Evaluated overall business performance over time.

Store Type Revenue by Year:

Store Type revenue by Year				
Total Revenue	Store Type			
Year of Sale	Flagship	Online	Outlet	Grand Total
2023	\$ 45,425.37	\$ 60,193.81	\$ 28,876.42	\$ 1,34,495.60
2024	\$ 2,46,165.05	\$2,63,449.25	\$1,19,641.48	\$ 6,29,255.79
2025	\$ 2,03,773.69	\$2,20,115.26	\$ 89,326.33	\$ 5,13,215.27
Grand Total	\$ 4,95,364.11	\$5,43,758.32	\$2,37,844.24	\$ 12,76,966.67



- Compared revenue generated by different store types across years.
- Identified the most profitable store type.
- Evaluated store-wise contribution to annual revenue.

7. Key Findings

- Electronics generated the highest revenue among product categories.
- North region recorded the highest overall revenue.
- Revenue showed a strong increase over the years, indicating business growth.
- Payment methods contributed differently to total revenue, helping identify customer purchasing preferences.
- Store type analysis highlighted variations in revenue generation across years.

8. Learning Outcomes

- Improved Excel dashboard development skills.
- Learned data cleaning and transformation techniques.
- Gained experience in business data analysis.
- Enhanced data visualization and reporting skills.

9. Conclusion

This project demonstrates the use of Microsoft Excel for data cleaning, transformation, analysis, and dashboard creation. The interactive dashboard provides valuable insights into sales performance and supports data-driven decision-making. Through this project, practical skills in Excel analytics, Pivot Tables, Pivot Charts, and dashboard design were strengthened.

Dashboard Screenshot:

